

|      |   |   |    |    |    |
|------|---|---|----|----|----|
| x    | 1 | 3 | 5  | 7  | 9  |
| f(x) | 0 | 6 | 18 | 29 | 42 |

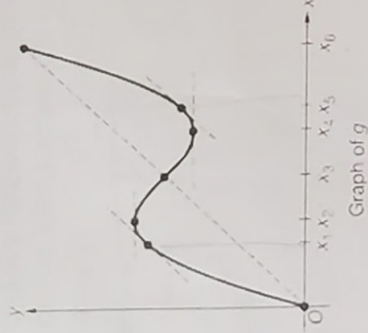
$$\frac{f(4)-f(1)}{8} = \frac{42-0}{8} = 5.25$$

This means no values are guaranteed

a) Zero b) One

c) Two d) Three

2. The function  $g$  shown in the diagram is continuous on the closed interval  $[0, x_6]$  and differentiable on the open interval  $(0, x_6)$ , where  $x_1, x_2, x_3, x_4, x_5$ , and  $x_6$  are points on the  $x$ -axis. Based on the graph what are all the values of  $x$  that satisfy the conclusion of the Mean Value Theorem applied to  $g$  on the closed interval  $[0, x_6]$ ? Explain.



a)  $x_3$  only

b)  $x_2$  and  $x_4$  only

c)  $x_1$  and  $x_5$  only

d)  $x_1, x_3$ , and  $x_5$  only

This is because the slope of the secant from  $(0, x_6)$  is the same slope as the points  $x_3$  and  $x_5$ .

3. The Mean Value Theorem can be applied to which of the following functions on the closed interval  $[-5, 5]$ ? Show work.

a)  $f(x) = \frac{1}{\sin x}$  Undefined / not cont

c)  $f(x) = \frac{x^2}{x^2-36}$  (not cont at  $x=6$  and  $x=-6$ )

b)  $f(x) = \frac{x-1}{|x-1|}$  Not defined / discontinuous from  $(-5, 5)$

d)  $f(x) = \frac{x^2}{x^2-4}$   $f(2)$  undefined so not possible

4. Which of the following functions of  $x$  is guaranteed by the Extreme Value Theorem to have an absolute maximum on the interval  $[0, 2\pi]$ ? Show work.

a)  $f(x) = \frac{1}{1+\sin x}$   $f(\frac{\pi}{2})$  is not defined

c)  $f(x) = \frac{x^2-2x\pi+\pi^2}{x-\pi}$  not defined at  $x=\pi$

b)  $f(x) = \frac{1}{x^2+\pi}$  cont over  $(0, 2\pi)$

d)  $f(x) = \frac{|x-\pi|}{x-\pi}$   $f(\pi)$  DNE so cannot apply EVT

5. Let  $g$  be the function given by  $g(x) = \sqrt{1 + \cos x}$ . Which of the following statements can be false on the interval  $\frac{\pi}{2} \leq x \leq \frac{7\pi}{4}$ ? Show work.

a) By the EVT, there is a value  $c$  such that  $g(c) \leq g(x)$  for  $\frac{\pi}{2} \leq x \leq \frac{7\pi}{4}$

b) By the EVT, there is a value  $c$  such that  $g(c) \geq g(x)$  for  $\frac{\pi}{2} \leq x \leq \frac{7\pi}{4}$  is possible because  $g$  cont over  $[\frac{\pi}{2}, \frac{7\pi}{4}]$

c) By the IVT, there is a value  $c$  such that  $g(c) = \frac{g(\frac{\pi}{2}) + g(\frac{7\pi}{4})}{2}$  yes as IVT guarantees a point between  $g(\frac{\pi}{2})$  and  $g(\frac{7\pi}{4})$

d) By the MVT, there is a value of  $c$  such that  $g'(c) = \frac{g(\frac{7\pi}{4}) - g(\frac{\pi}{2})}{\frac{7\pi}{4} - \frac{\pi}{2}}$  false as  $g$  not differentiable over  $[\frac{\pi}{2}, \frac{7\pi}{4}]$

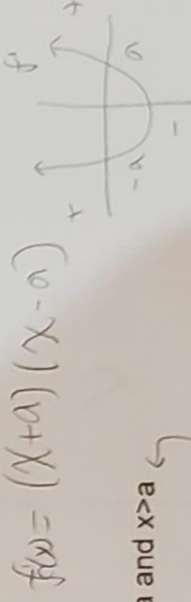
6. Let  $f$  be the function with derivative given by  $f'(x) = x^2 - a^2$ , where  $a$  is a positive constant, which of the following statements is true? Show work.

a)  $f$  is decreasing for  $-a < x < a$  because  $f'(x) < 0$  for  $-a < x < a$

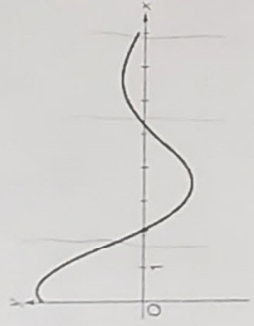
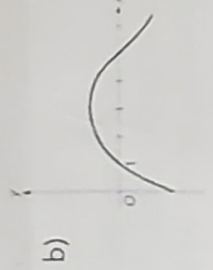
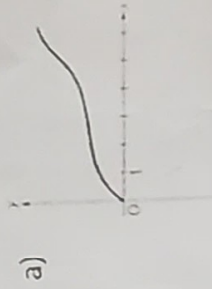
b)  $f$  is decreasing for  $x < -a$  and  $x > a$  because  $f'(x) < 0$  for  $x < -a$  and  $x > a$

c)  $f$  is decreasing for  $x < 0$  because  $f'(x) < 0$  for  $x < 0$ .

d)  $f$  is decreasing for  $x < 0$  because  $f''(x) < 0$  for  $x < 0$ .



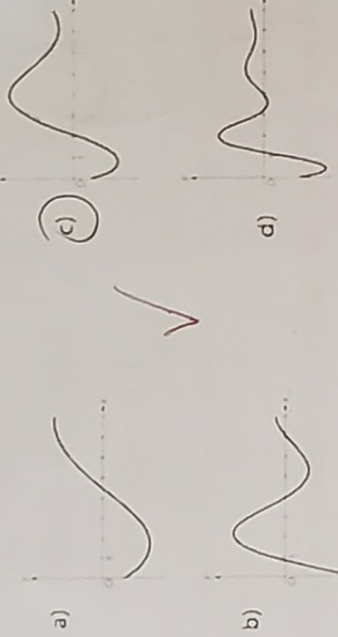
points of inflection on this interval. Which of the following could be the graph of  $f'$ , the derivative of  $f$ ? Write down at least one reason.



Graph of  $f''$

8. The graph of  $f''$ , the second derivative of the function  $f$  is shown above on the interval  $0 \leq x \leq 8$ . Which of the following could be the graph of  $f$ ? Write down at least one reason.

Concavity switches 7 times and so the graph of  $f$  crosses the x-axis 7 times



9. Let  $g$  be the function defined by  $g(x) = |x| - 3|x + 1|$ . What is the absolute maximum value of  $g$  on the closed interval  $[-2, 2]$ ? Show work.

- (a) 1      b) -1      c) -3      d) -7

$$g(-2) = 2 - 3 = -1$$

$$g(2) = -7$$

$$g(-1) = 1$$

10. At what values of  $x$  does the graph  $y = x^2 e^{-2x}$  have a point of inflection? Show work.

- a)  $x = -2$  and  $x = 0$

- c)  $x = -2 - \sqrt{2}$  and  $x = -2 + \sqrt{2}$

- b)  $x = 1$  and  $x = 0$

- d)  $x = 1 - \frac{\sqrt{2}}{2}$  and  $x = 1 + \frac{\sqrt{2}}{2}$

$$g(x) = x^2 e^{-2x}$$

$$g'(x) = 2x e^{-2x} - 2x^2 e^{-2x}$$

$$g''(x) = 2e^{-2x} - 4x e^{-2x} - 4x e^{-2x} + 4x^2 e^{-2x}$$

$$g''(x) = 2e^{-2x} (1 - 4x + 2x^2)$$

Roots are  $1 \pm \frac{\sqrt{2}}{2}$

### Part 2: Problem Solving

1. Let  $f(x) = (x-1)^2$ ,  $g(x) = e^{-2x}$  and  $h(x) = 1 + \ln(1-2x)$ .

a) Find the linearization of  $f$ ,  $g$  and  $h$  at  $a = 0$ . What do you notice? Explain why this happens? (3)

$$f(x) = (x-1)^2$$

$$f'(x) = 2(x-1)$$

$$f(0) = 1$$

$$g(x) = e^{-2x}$$

$$g'(x) = -2e^{-2x}$$

$$g(0) = 1$$

$$g'(0) = -2$$

$$h(x) = 1 + \ln(1-2x)$$

$$h'(x) = \frac{-2}{1-2x}$$

$$h'(0) = -2$$

$$h(0) = 1$$

$$h'(0) = -2$$

$$h(0) = 1$$

$$L(x) = 1 - 2x$$

$$(1 - 0.1)^2 \text{ vs } (1 - 2(0.1))$$

$$0.81 \quad 0.64$$

opt by 0.1

Best choice

$$L(x) = 1 - 2x$$

$$0.81 \quad 0.64$$

$$\text{opt by } 0.1875$$

show

by

$$L(x) = 1 - 2x$$

$$L(0.1) = 0.80 \quad h(0.1) = 0.77$$

opt by max then 0.03

Pick 2 or 3.

2. The square sheet of the cardboard has a side length of 40 inches. Four squares with side length  $x$  and two rectangular regions are discarded to create a rectangular box with a lid. Determine the critical points. Use second derivative test to determine the maximum volume of the box.

$$\frac{3.5}{(4)}$$

$$V = (20 - x)(40 - 2x)(x)$$

$$V = (800 - 40x - 40x + 2x^2)(x)$$

$$V = 800x - 80x^2 + 2x^3$$

$$V' = 800 - 160x + 6x^2 \quad \text{use quadratic formula}$$

$$V' = 2(400 - 80x + 3x^2)$$

$$V'' = 2(-80 + 6x)$$

$$V''(0) = -80 \text{ so since negative}$$

Test with second derivative

volume

derivative

testing second derivative

Downward.

$$80 \pm \sqrt{80^2 - 4(3)(400)}$$

$$80 \pm \sqrt{1600}$$

$$\frac{80 \pm 40}{6}$$

$$\frac{80}{6} \text{ or } \frac{120}{6}$$

$$x = 20 \text{ or } x = 20$$

$$x = 20 \text{ is invalid so } x = \frac{40}{3}$$

3. A coned shaped paper drinking cup is to be made to hold  $27\text{cm}^3$  of water. Find the height and radius of the cup that will use the smallest amount of paper. ( $V = \frac{1}{3}\pi r^2 h$   $S = \pi r \sqrt{r^2 + h^2}$ )

$$27\text{cm}^3 = V$$

$$\frac{81}{\pi r^2} = h$$

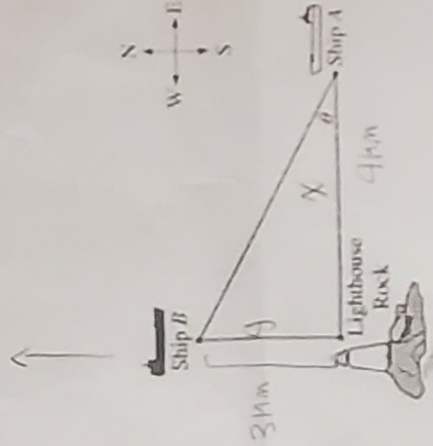
$$S = \pi r \sqrt{r^2 + h^2}$$

$$S = \pi r \sqrt{\frac{81}{r^2} + r^2}$$

$$S' =$$

(4)

is 3 km away from the lighthouse rock.



$$\frac{dy}{dt} = 10 \text{ km/h} \quad (4)$$

$$\frac{dx}{dt} = -15 \text{ km/h}$$

$$y = 3$$

$$x = 4$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\frac{d\theta}{dt} = \frac{1}{\left(1 + \left(\frac{y}{x}\right)^2\right)} \left(\frac{\frac{dy}{dt} - y \frac{dx}{dt}}{x^2}\right)$$

$$= \left(\frac{1}{1 + \left(\frac{3}{4}\right)^2}\right) \left(\frac{4(10) - (3)(-15)}{16}\right)$$

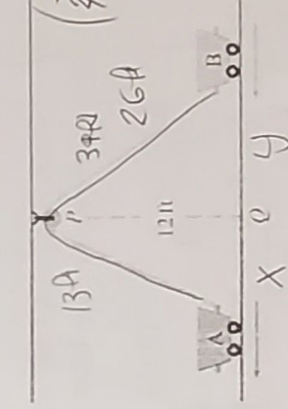
$$\frac{d\theta}{dt}$$

$$= 3.4104/\text{h}$$

Maths sense as thing align should be getting larger

5. Two carts A and B, are connected by a rope 39 ft long that passes over a pulley P as shown in the diagram. Point Q is on the floor 12 ft directly beneath P and between the carts. Cart A is being pulled away from Q at a speed of 2 ft/s. How fast is cart B moving toward Q at the instant when cart A is 5 ft from Q.

$$\sqrt{12^2 + x^2} + \sqrt{y^2 + 14^2} = 39 \quad \text{Take implicit (4)}$$



$$\left(\frac{1}{\sqrt{144 + x^2}} \cdot x \cdot \frac{dx}{dt}\right) + \left(\frac{1}{\sqrt{y^2 + 14^2}} \cdot 2y\right) \cdot \frac{dy}{dt} = 0 \quad \text{Solving for } \frac{dy}{dt}$$

$$\left(\frac{x}{\sqrt{144 + x^2}} \cdot \frac{dx}{dt}\right) + \left(\frac{y}{\sqrt{y^2 + 14^2}} \cdot \frac{dy}{dt}\right) = 0$$

$$\left(\frac{5}{\sqrt{144 + 25}} \cdot 2\right) + \frac{2\sqrt{133}}{26} \cdot \frac{dy}{dt} = 0$$

$$y = 5 \quad y = 2\sqrt{133}$$

$$\left(\frac{10}{13}\right) + \frac{2\sqrt{133}}{26} \cdot \frac{dy}{dt} = 0$$

-10 26 values sense as y value should be decreasing

5. Use Newton's method to approximate the negative root of  $e^x = 4 - x^2$ , correct to six decimal places.

$$\text{Formula: } x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$\text{Start guess at } -1.75 \quad f(x) = 4 - x^2 - e^x$$

$$x_{n+1} = x_n - \frac{4 - x^2 - e^x}{-2x - e^x}$$

$$x_{n+1} = -1.75 - \frac{4 - (-1.75)^2 - e^{-1.75}}{-2(-1.75) - e^{-1.75}}$$

$$x_{n+1} = -4 - \frac{4 - (-4)^2 - e^{-4}}{-2(-4) - e^{-4}} \rightarrow 1.058$$

So lets start at -4

